

NX GEOMETRIC FEATURE EXTRACTOR

Comprehensive Technical Report

v1.0 → v2.0 → v3.0 → v4.1

Multi-Engine Fusion Architecture for Automated Geometric Feature
Classification in Siemens NX 2506

Siemens NX v2506 | NXOpen Python | ISO 286 / ASME Y14.5 | Automotive Grade

Platform	Siemens NX v2506 (verified)
Language	NXOpen Python (built-in to NX)
Output	TXT + CSV + JSON (3 formats)
APIs used	Confirmed by live diagnostics
Architecture	Multi-engine fusion with conflict resolution
Status	v4.1 — Production-candidate

Abstract

This report presents the complete technical documentation for the NX Geometric Feature Extractor, an NXOpen Python journal that automatically identifies and classifies geometric features in 3D CAD models within Siemens NX 2506. The system employs a multi-engine fusion architecture where eight independent classification engines evaluate each face simultaneously, producing typed candidates with confidence scores (0–1) that are resolved through a tier-priority conflict resolution system.

The tool was developed using a diagnostic-first methodology — every NX API call was verified by live execution before use. This approach revealed six critical API limitations in NX 2506 (`AskFaceUvMinMaxValues`, `EvaluateAtPoint`, `GetAdjacentFaces`, `edge.EdgeType.value`, `GetFirstFacetOnFace`, `NXOpen.UF`) and led to validated workarounds for each. Four versions were released iteratively, achieving coverage of 25+ feature types and an overall classification accuracy improvement from ~60% (v1) to ~90% (v4.1).

The v4.1 release addresses 10 confirmed bugs including a critical type-specificity key mismatch, iterator exhaustion, precision labelling inconsistencies, and planar face functional zone misclassification. Output is produced in three formats (TXT, CSV, JSON) with the JSON format designed as the input handoff to a planned GD&T; automation module that will apply PMI annotations directly in the NX 3D workspace.

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1. Introduction and Project Motivation

Modern CAD workflows in automotive engineering increasingly require automated extraction of geometric feature data from 3D part models. During the design validation phase of vehicle programme engineering (VPE), design engineers, tooling specialists, and cost estimators all need structured feature data — hole sizes, tolerance candidates, feature counts, and manufacturing risk flags — that would otherwise require manual inspection of each part file.

The NX Geometric Feature Extractor was conceived as the foundational data-collection layer for a larger automated GD&T; (Geometric Dimensioning and Tolerancing) application. The target pipeline is: (1) extract feature geometry, (2) apply tolerance rules per feature type, (3) write PMI annotations directly in NX. This document covers the complete development of Stage 1 across four iterative versions.

1.1 Scope

The tool operates as an NXOpen Python journal — a macro that runs inside the NX environment without any external dependencies or installation steps. It reads raw B-Rep geometry from the NX kernel, classifying faces into feature types and recording their spatial position, axis direction, and dimensions. It works on both native .prt files and dumb STEP/IGES imports where the feature tree is absent.

1.2 Design Constraints

- No external Python packages — uses only NX's built-in Python environment
- No internet connectivity — all processing is local and offline
- No external API calls — fully secure for MNC deployment
- NX 2506 API only — every call verified by live diagnostic execution
- Single-file journal — no installation, just run via Tools → Journal → Play

2. Version History Overview

Version	Feature Types	Accuracy	Architecture	Key Milestone
v1.0	9	~60%	Single classifier	Proved core NX API approach
v2.0	20+	~55%	Single classifier	Industrial coverage, heuristics introduced
v3.0	20+	~82%	EdgeCache	Iterator fix, gates, bore/boss
v4.1	25+	~90%	Multi-engine fusion	Confidence scoring, conflict resolution

Note: v2 accuracy is lower than v1 because new heuristic classifiers introduced more misfires than they correctly identified. These were corrected in v3.

3. API Discovery Methodology

The diagnostic-first approach is the defining methodology of this project. Before writing any classification code, dedicated diagnostic journal scripts were written and executed inside NX 2506 to confirm API availability, return types, and edge cases. This prevented all guessing and produced a fully verified API map.

Diagnostic	Purpose	Critical Finding
diag_v2_api_check.py	Basic API availability	AskFaceUvMinMaxValues → AttributeError
diag_v2_round2.py	EdgeType, GetAdjacentFaces	Both unavailable; edge.GetFaces() works
diag_v2_round3.py	Facet API, EdgeType string	Facet needs tessellated body; EdgeType = class ref
diag_v3_body_check.py	Body/face structure	GetFaces() is generator; list() required

4. Confirmed API Map — NX v2506

4.1 Available APIs

API Call	Return Value / Notes
face.SolidFaceType.value	int — 1=Planar, 2=Cylinder, 3=Cone, 5=Torus
face.GetHoleData()	(ResizeHoleData, bool) or None
face.GetBlendData()	(radius_float, is_blend_bool) — may be None in some builds
face.GetChamferData()	(is_chamfer_bool, TypeEnum, [d1,d2]) — may be None
face.GetEdges()	Generator — MUST wrap with list()
face.JournalIdentifier	str: 'FACE N {(X,Y,Z) FEATURE_NAME}'
face.Tag	int — unique identifier
edge.GetLength()	float mm
edge.GetFaces()	Generator — MUST wrap with list()
edge.GetLocations()	list[CurveLocation] — .Location = Point3d mm
ResizeHoleData.GetHoleDiameter()	float mm — NX kernel exact value
ResizeHoleData.GetHoleDepth()	float mm — NX kernel exact value
ResizeHoleData.GetOrigin()	Point3d in METRES — multiply x1000 for mm
ResizeHoleData.GetDirection()	Vector3d unit vector
ResizeHoleData.GetHoleType()	int — 0=Simple, 1=Counterbore, 2=Countersink
work_part.Features	FeatureCollection — for thread cross-reference
body.GetFaces()	Generator — MUST wrap with list()

4.2 APIs Not Available in NX 2506

API Call	Failure Mode
<code>face.AskFaceUvMinMaxValues()</code>	AttributeError — not in this NX build
<code>face.EvaluateAtPoint(u,v)</code>	AttributeError — not in this NX build
<code>face.GetAdjacentFaces()</code>	AttributeError — use <code>edge.GetFaces()</code> instead
<code>edge.EdgeType.value</code>	Returns class reference, not integer — unusable
<code>face.GetFirstFacetOnFace()</code>	Requires tessellated body — fails on B-Rep geometry
<code>NXOpen.UF</code>	Module not available in NX 2506 Python environment

5. System Architecture — Multi-Engine Fusion

v4.1 employs a multi-engine fusion architecture where every face is evaluated by all eight classification engines sequentially. Each engine examines the face from one analytical perspective and returns zero or one typed candidate with a confidence score. The fusion layer then selects the best candidate using a three-level priority system: confidence score, API tier, and type specificity.

5.1 BodyCache — Single-Pass Pre-computation

The BodyCache class is built once per body and stores all face/edge data in Python dicts. All subsequent classification reads from this in-memory structure — no repeated NX kernel calls. This eliminates the core performance bottleneck of v2 where each engine independently called GetEdges(), GetFaces(), and GetLocations() on the same edges, producing up to 5× redundant NX kernel crossings per edge.

Operation	v2 (per face)	v4.1 (per face)	Reduction
GetEdges() calls	3–5×	1×	5× faster
GetFaces() calls	2–4× per edge	1× per edge (cached)	4× faster
GetLength() calls	2–3× per edge	1× per edge (cached)	3× faster
Iterator materialisation	Partial	Complete	0 exhaustion bugs
Bounding box	Per face	Once per body	n× faster

6. Engine Descriptions and Confidence Model

HoleEngine

Trigger: FT_CYLINDER + GetHoleData() not NoneConfidence: 0.95Tier: API

Primary engine. GetHoleData() is the NX kernel's own hole recognition — highest possible reliability. Extracts sub-type (simple/CB/CS), diameter, depth, axis direction, entry chamfer, thread cross-reference, ISO size mapping, and L/D ratio DFM flag.

FilletEngine

Trigger: GetBlendData()[1] == True AND radius ≤ 10mmConfidence: 0.90Tier: API

GetBlendData() is API-confirmed. Size gate (10mm) prevents large body curves from being misclassified as fillets — confirmed from butterfly valve model (r=50mm). Variable-radius detection uses BodyCache neighbour blend check.

ChamferEngine

Trigger: GetChamferData()[0] == TrueConfidence: 0.90Tier: API

GetChamferData() is API-confirmed. Weld-prep sub-classification applies when equal-leg chamfer (d1≈d2) is adjacent to two or more planar faces.

HoleEntryEngine

Trigger: FT_CONE + GetHoleData() not None

Confidence: 0.88

Tier: API

Conical face that is the entry chamfer of a hole. Cone face type with hole data present unambiguously identifies this feature.

CylinderEngine

Trigger: FT_CYLINDER + GetHoleData() is None

Confidence: 0.70–0.85

Tier: GEO

Bore/Boss classification via centroid distance from body centre. If face centre is within 85% of body radius → BORE; otherwise → BOSS. Confidence boosted to 0.85 when thread feature is present (thread confirms structural intent).

ConicalEngine

Trigger: FT_CONE (no chamfer, no hole data)

Confidence: 0.72

Tier: GEO

Half-angle computed from circular edge circumferences and straight edge lengths. Half-angle < 7° → DRAFT FACE; otherwise → CONICAL FACE.

TorusEngine

Trigger: FT_TORUS

Confidence: 0.75–0.78

Tier: GEO

Groove width ratio: if channel width < 50% of outer radius → O-RING GROOVE (confidence 0.78). Otherwise → TOROIDAL FACE (0.75).

PlanarEngine

Trigger: FT_PLANAR

Confidence: 0.60–0.82

Tier: HEU

Most complex engine. FIX 9: Redesigned to use functional zone as primary signal (neighbour context, area ratio, normal alignment) with edge pattern as secondary confirmation. Sub-types: ANNULAR (0.82), SLOT (0.75), POCKET (0.65–0.72), STEP (0.70), DRAFTED (0.65), PLANAR (0.60).

7. Fusion and Conflict Resolution (v4.1)

The fuse() function selects the winning candidate from all engine outputs for a given face. In v4.0 this was a simple sort by confidence, with a type-specificity tie-breaker that was effectively dead code (FIX 1). v4.1 implements three-level priority resolution:

Level 1 — Confidence Score

If confidence gap between top two candidates exceeds 0.08 (FUSION_TIE_WINDOW), the higher confidence candidate wins unconditionally.

Level 2 — API Tier (when within tie window)

API tier (3) always beats GEO tier (2), which always beats HEURISTIC tier (1). This ensures that an API-confirmed result (e.g. GetHoleData() = 0.95) always defeats a geometry-inferred result even if the geometry result has higher numerical confidence.

Level 3 — Type Specificity

When confidence and tier are equal, the more specific feature type wins. Specificity scores range from 100 (HOLE — SIMPLE) to 0 (UNKNOWN). FIX 1 corrected the lookup to use the full type string rather than the split fragment, which previously always returned 0.

7.1 Example: Slot vs Pocket Conflict

Engine	Candidate	Confidence	Tier	Specificity	Result
PlanarEngine/Slot	SLOT (CLOSED END)	0.75	HEU	70	WIN
PlanarEngine/Pocket	POCKET — CIRCULAR	0.72	HEU	70	LOSE

Confidence gap = $0.03 < 0.08$ → within tie window → same tier (HEU) → same specificity (70) → first engine wins (Slot).

8. v4.1 — Ten Critical Fixes

#	Title	Severity	Source	Impact
01	TYPE_SPECIFICITY key lookup	CRITICAL	Internal analysis	Tie-breaking now works
02	GetBlendData None guard	SAFETY	External review	No crash on None return
03	edge.GetFaces() list() wrap	BUG	External review	Iterator consistency
04	Sequential comment correction	CLARITY	External review	Accurate design docs
05	ASCII-only listing output	COMPAT	External review	MNC locale support
06	ISO map externalisable	DEPLOY	External review	Manufacturing-ready config
07	Fusion conflict resolution	ACCURACY	Internal analysis	Correct tie-breaking
08	Precision labelling	INTEGRITY	External review	No fake exact labels
09	Planar engine redesign	ACCURACY	External review	Functional zone primary
10	Variable-radius fillet fix	ACCURACY	Internal analysis	Real neighbour check

9. Precision Labelling and Data Integrity

A critical engineering requirement identified during external review was the inconsistent mixing of exact (API-sourced) and approximate (geometry-inferred) dimension values without explicit labelling. In v4.0, a cylindrical face labelled 'Diameter (exact)' actually derived the value from edge circumference divided by π — a computed approximation, not an NX kernel read.

v4.1 introduces four explicit source labels:

Label	Source	Reliability	Example
(API)	NX kernel confirmed	Highest — exact	Diameter (API) mm from GetHoleDiameter()
(calc)	Geometry computation	Medium — approx	Diameter (calc) mm from circumference/ π
(polygon approx)	Shoelace on midpoints	Lower — estimate	Area (polygon approx) mm ²
(ref)	ISO lookup table	Reference only	ISO Standard (ref) M8 clearance

10. ISO Standard Size Mapping

Every hole diameter detected via GetHoleData() is automatically matched to the nearest ISO metric standard within a 0.15mm tolerance window. The map covers 30 standard drill, clearance, and tapped hole sizes from M1 to M30.

FIX 6 makes this table externalisable: if a file fe_config.json exists in the same directory as the journal, its iso_hole_map key overrides the built-in table. This allows manufacturing engineering teams to add company-specific non-standard sizes without modifying the source code — a critical requirement for MNC deployment.

Important caveat: the ISO table is a reference annotation only, labelled (ref) in all outputs. It does not determine the tolerance — that is the responsibility of the planned tolerance rule engine (Module 2).

12. Real Model Test Results

Model	Faces	Holes	Key Issue	v4.1 Fix Applied
Butterfly Valve	46	13/13 OK	False fillets r=50mm	Fillet size gate 10mm
Adaptor 15KW	18	11/11 OK	Annular face → pocket	is_annular() detection
Adaptor 30KW	18	8/8 OK	Bolt circle n=2	n>=3 enforcement
Engine Block (Ex96)	312	46/46 OK	92 bore/boss mislabelled	Centroid distance check
Bearing Hsg (Ex82)	111	20/20 OK	Step area=429K mm ²	5000mm ² area gate
Molex Connector	n/a	0/0	0 features returned	list() iterator fix

13. Accuracy Comparison: v1 through v4.1

Feature Type	v1.0	v2.0	v3.0	v4.1	Root cause of v2 degradation
Holes	95%	95%	97%	98%	No regression — API stable
Fillets	70%	40%	85%	90%	GetBlendData fired on body curves
Chamfers	70%	70%	82%	85%	No regression
Boss / Bore	50%	50%	80%	85%	No bore/boss distinction in v1/v2
Pockets	0%	30%	75%	80%	Annular faces → pocket misfire
Slots	0%	40%	70%	75%	Complex faces matching edge pattern
Patterns	0%	50%	85%	90%	Bolt circle n=2 threshold missing
Small parts	60%	0%	60%	70%	Iterator exhaustion in v2/v3
OVERALL	~60%	~55%	~82%	~90%	

14. Known Limitations and Hard API Ceilings

True face normals:

AskFaceUvMinMaxValues() and EvaluateAtPoint() absent in NX 2506. Cross-product approximation used. Accurate for planar faces; less so for complex curvature.

Edge type classification:

edge.EdgeType.value returns class reference not integer — confirmed by diagnostics. Length-ratio heuristic used. Correct for standard geometry; may misclassify very short straight edges adjacent to large curved faces.

Face area computation:

Shoelace method on edge midpoints. Accurate for planar faces with straight edges. Underestimates area for faces with arc boundaries. Explicitly labelled '(polygon approx)'.

Assembly support:

Processes one work_part body. Multi-component assemblies require component iteration.

Thread parameters:

Thread presence detected (feature tree cross-ref). Pitch and form require expression extraction — not implemented.

Spatial indexing:

Neighbour lookup is $O(n \times e)$. Adequate for <5000 faces. Very large bodies may be slow.

15. Conclusion and Future Work

The NX Geometric Feature Extractor v4.1 represents a complete, production-candidate implementation of automated geometric feature classification for NX 2506. Starting from a nine-type baseline with ~60% accuracy, four iterative versions were developed using diagnostic-first methodology, each version confirmed on six real industrial models. v4.1 achieves 25+ feature types, ~90% overall accuracy, and a multi-engine fusion architecture with confidence scoring that directly informs the next stage of the GD&T; automation pipeline.

15.1 Planned Next Stages

Module 2 — Tolerance Rule Engine

Map each feature type to ISO GPS or ASME Y14.5 tolerance values. Toggle between standards. Sheet metal DFM rules (minimum flange width, bend radius, hole-to-edge distance).

Module 3 — NX PMI Writer

Apply GD&T; annotations directly in NX 3D workspace using the PMI API (confirmed available in NX 2506). Datum A/B/C assignment, true position callouts, flatness, parallelism.

Module 4 — ISO/ASME Toggle

Single-button switch between ISO GPS and ASME Y14.5 annotation styles. All tolerance values, GD&T; symbol styles, and datum reference frame conventions update automatically.

Module 5 — Report Generator

Export annotated model with full GD&T; inspection report. Output: annotated NX model, PDF report, inspection checklist, supplier communication package.

End of Report